

LINEAR TRANSFORMATIONS

DEFINITION

Let V, W be two vector spaces. A function $L : V \rightarrow W$ is said to be linear (or a linear transformation) if for all $\vec{v}, \vec{w} \in V$ and $c \in \mathbb{R}$

1. $L(\vec{v} + \vec{w}) = L(\vec{v}) + L(\vec{w})$ (Additivity)
2. $L(c\vec{v}) = cL(\vec{v})$ (Homogeneity)
- THEOREM 4.2**
- Let $L : V \rightarrow W$ be a function between vector spaces. Then, the following are equivalent
1. L is linear
2. $L(\vec{u} + c\vec{v}) = L(\vec{u}) + cL(\vec{v})$ for all $\vec{u}, \vec{v} \in V$ and all $c \in \mathbb{R}^n$
3. $L(a\vec{u} + b\vec{v}) = aL(\vec{u}) + bL(\vec{v})$ for all $\vec{u}, \vec{v} \in V$ and all $c \in \mathbb{R}^n$

DERIVATIVES

SINGLE VARIABLE

Recall the definition of a derivative on for a function $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

This can also be written as

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - f'(a)h}{h} = 0$$

THEOREM 7.10

The function $f : I \rightarrow \mathbb{R}^n$ is differentiable at some $a \in I$, where I is an open interval, iff there exists a linear transformation $L : \mathbb{R} \rightarrow \mathbb{R}^n$ such that

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - L(h)}{h} = 0$$

Furthermore, when such a linear transformation exists, it is unique and

$$L(h) = f'(a)h$$

DIRECTIONAL DERIVATIVES

Let $\mathbf{a}$ be a point in an open subset U of $\mathbb{R}^n$ and $F : U \rightarrow \mathbb{R}^m$ be a function, and $\mathbf{0} \neq \mathbf{v} \in \mathbb{R}^n$. The directional derivative of F with respect to $\mathbf{v}$ at $\mathbf{a}$ is defined by

$$D_{\mathbf{v}}F(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{F(\mathbf{a} + h\mathbf{v}) - F(\mathbf{a})}{h}$$

DERIVATIVES, DIFFERENTIALS, AND THE JACOBIAN

Let $\mathbf{a}$ be a point in an open subset U of $\mathbb{R}^n$. We say $F : U \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{a}$ if there exists a linear transformation $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - L(\mathbf{h})}{\|\mathbf{h}\|} = 0$$

Let $\mathbf{x}$ be a point in an open subset U of $\mathbb{R}^n$. If $F = (F_1, \dots, F_m) : U \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{x}$, then

$$D_{\mathbf{v}}F(\mathbf{x}) = dF_{\mathbf{x}}(\mathbf{v}) = F'(\mathbf{x})\mathbf{v}$$

Furthermore, then (i, j) entry of $F'(\mathbf{x})$ is $\frac{\partial F_i}{\partial x_j}(\mathbf{x})$. In other words,

$$F'(\mathbf{x}) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1}(\mathbf{x}) & \frac{\partial F_1}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial F_1}{\partial x_n}(\mathbf{x}) \\ \frac{\partial F_2}{\partial x_1}(\mathbf{x}) & \frac{\partial F_2}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial F_2}{\partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1}(\mathbf{x}) & \frac{\partial F_m}{\partial x_2}(\mathbf{x}) & \cdots & \frac{\partial F_m}{\partial x_n}(\mathbf{x}) \end{pmatrix}$$

The gradient of a function $f : U \rightarrow \mathbb{R}$, where U is an open subset of $\mathbb{R}^n$ is the function

$$\nabla f : U \rightarrow \mathbb{R} \text{ defined by } \nabla f(\mathbf{a}) = (D_1f(\mathbf{a}), \dots, D_nf(\mathbf{a}))$$

This is a consequence of the fact that the directional derivative of a function at $\mathbf{a}$ relative to $\mathbf{v} \in \mathbb{R}^n$ can be computed via

$$D_{\mathbf{v}}F(\mathbf{a}) = dF_{\mathbf{a}}(\mathbf{v}) = J_{\mathbf{a}}\mathbf{v}$$

Let U be an open subset of $\mathbb{R}^n$, and $F : U \rightarrow \mathbb{R}^m$ be differentiable. Suppose $\mathbf{a} \in U$. Then, the approximation

$$F(\mathbf{x}) \approx F(\mathbf{a}) + dF_{\mathbf{a}}(\mathbf{x} - \mathbf{a})$$

This is the tangent plane approximation of F near $\mathbf{a}$

THE CHAIN RULE

Suppose U and V are open subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Suppose $F : U \rightarrow \mathbb{R}^m$ and $G : V \rightarrow \mathbb{R}^k$ are differentiable at points $\mathbf{a} \in U$, and $F(\mathbf{a}) \in V$, respectively. Assume $F(U) \subseteq V$. Then, the composition $H = G \circ F$ is differentiable at $\mathbf{a}$ and

$$dH_{\mathbf{a}} = dG_{F(\mathbf{a})} \circ dF_{\mathbf{a}}$$

FUNCTION EXTREMA

QUADRATIC FORMS & SECOND PARTIALS TEST

Let $f : U \rightarrow \mathbb{R}$ be a twice continuously differentiable, where U is an open subset of $\mathbb{R}^2$. Suppose $\mathbf{a} \in U$ is a critical point of f . Let

$$\Delta = \frac{\partial^2 f}{\partial x^2}(\mathbf{a}) \cdot \frac{\partial^2 f}{\partial y^2}(\mathbf{a}) - \left(\frac{\partial^2 f}{\partial x \partial y}(\mathbf{a}) \right)^2$$

Then f has

- A local minimum at $\mathbf{a}$ if $\Delta > 0$ and $\frac{\partial^2 f}{\partial x^2}(\mathbf{a}) > 0$
- A local maximum at $\mathbf{a}$ if $\Delta > 0$ and $\frac{\partial^2 f}{\partial x^2}(\mathbf{a}) < 0$
- A saddle point at $\mathbf{a}$ if $\Delta < 0$
- Indeterminate if $\Delta = 0$

Eigenvalues of $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$

Given a quadratic form q on $\mathbb{R}^n$, the maximum and minimum value of this quadratic form on the unit sphere $\{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = 1\}$ are the largest and smallest eigenvalues of the matrix A associated with q .

THE HESSIAN

Let $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ be a quadratic form where A is symmetric

1. q positive definite if and only if $\Delta_k > 0$ for all k
2. q is negative definite if and only if $(-1)^k \Delta_k > 0$ for all k (they alternate in sign with $k = 1$ being negative)
3. Assume $\det A \neq 0$. The quadratic form q is nondefinite if and only if neither of the previous two conditions is satisfied

Theorem 10.8

Let U be an open subset of $\mathbb{R}^n$. Suppose $f : U \rightarrow \mathbb{R}$ is C^2 , and let $\mathbf{a} \in U$ be a critical point of f

1. If $H_f(\mathbf{a})$ is positive-definite, then f has a local minimum at $\mathbf{a}$
2. If $H_f(\mathbf{a})$ is negative-definite, then f has a local maximum at $\mathbf{a}$
3. If $H_f(\mathbf{a})$ is nondefinite, then f has a saddle point at $\mathbf{a}$

INTEGRATION

CHANGE OF VARIABLES

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a C^1 function given by $T(u, v) = (x(u, v), y(u, v))$. Suppose D and $T(D)$ are elementary regions in the uv and xy planes, respectively. Assume f is integrable over $T(D)$, then

$$\iint_{T(D)} f(x, y) \, dA = \iint_{T(D)} f(u, v) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, dA$$

For every point $P \in \mathbb{R}^3$ we assign a triple (r, θ, z) called the cylindrical coordinates of P where (r, θ) are the polar coordinates of the point (x, y) Thus the conversion is given buy

$$x = r \cos(\theta) \qquad y = r \sin(\theta) \qquad z = z$$

Similarly, the spherical coordinates of P are given by a triple (ρ, φ, θ) where ρ is the distance of P from the origin and φ and θ and the angles made with different planes The conversions are

$$x = \rho \sin(\theta) \cos(\varphi) \qquad y = \rho \sin(\theta) \sin(\varphi) \qquad z = \rho \cos(\varphi)$$

And $r = \rho \sin(\theta)$. The Jacobian coefficients of these are Cylindrical:

$$dx \, dy \, dz = r \, dr \, d\theta \, dz$$

Spherical:

$$dx \, dy \, dz = \rho^2 \sin(\varphi) \, d\rho \, d\varphi \, d\theta$$

The surface given by $z = f(x, y)$ (i.e. the graph of a function $f(x, y)$) can be parametrized with

$$X(x, y) = (x, y, f(x, y)) \quad \text{where} \quad X_x \times X_y = (-f_x, -f_y, 1)$$

LINE & SURFACE INTEGRALS

Suppose $f : [a, b] \rightarrow \mathbb{R}^3$ is a simple C^1 curve. The arc length of this curve is given by

$$\int_a^b \|f'(t)\| \, dt$$

Consider some curve C given by a parametrization $\mathbf{x} : [a, b] \rightarrow \mathbb{R}^n$. The scalar line integral of a real valued function f over this curve is given by

$$\int_C f \, ds = \int_a^b f \circ \mathbf{x}(t) \, \|\mathbf{x}'(t)\| \, dt$$

Let $D \subseteq \mathbb{R}^n$ that contains the image of a path given by $\mathbf{x} : [a, b] \rightarrow \mathbb{R}^n$. The vector line integral of a vector field $F : D \rightarrow \mathbb{R}^n$ over $\mathbf{x}$ is given by

$$\int_{\mathbf{x}} \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F} \circ \mathbf{x}(t) \cdot \mathbf{x}'(t) \, dt$$

Let $D \subseteq \mathbb{R}^2$ be a bounded region. Let $X : D \rightarrow \mathbb{R}^3$ be a piecewise smooth parametrized surface. Let f be a continuous real-valued function whose domain contains $S = X(D)$. Then the scalar surface integral of f along X is given by

$$\iint_X f \, dS = \iint_D f \circ X(s, t) \, \|\mathbf{N}(s, t)\| \, dA = \iint_D f \circ X(s, t) \, \|\mathbf{X}_s \times \mathbf{X}_t\| \, dA$$

Let $D \subseteq \mathbb{R}^2$ be a bounded region. Let $X : D \rightarrow \mathbb{R}^3$ be a piecewise smooth parametrized surface. Let $\mathbf{F}(x, y, z)$ be a continuous real-valued function whose domain contains $S = X(D)$. Then the vector surface integral of $\mathbf{F}$ along X is given by

$$\iint_X \mathbf{F} \, d\mathbf{S} = \iint_D \mathbf{F} \circ X(s, t) \cdot \mathbf{N}(s, t) \, dA = \iint_D \mathbf{F} \circ X(s, t) \cdot (\mathbf{X}_s \times \mathbf{X}_t) \, dA$$

GREEN'S, STOKES'S, AND GAUSS' THEOREMS

Let D be a closed bounded region in $\mathbb{R}^2$, whose boundary ∂D consists of finitely many simple, closed, piecewise continuously differentiable curves. Suppose ∂D is oriented in such a way that D lies on the left as one traverses ∂D (it's counter-clockwise). Let $F = M\mathbf{i} + N\mathbf{j}$ be a continuously differentiable vector field on D . Then

$$\int_{\partial D} M \, dx + N \, dy = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \, dA$$

Let S be a bounded, piecewise smooth, oriented surface in $\mathbb{R}^3$. Suppose ∂S consists of finitely many piecewise continuously differentiable, simple, closed curves each of which is oriented consistently with S . Let $\mathbf{F}$ be a continuously differentiable vector field whose domain includes S . Then

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{s}$$

Let D be a bounded region whose boundary ∂D consists of finitely many piecewise smooth closed orientable surfaces, each of which is oriented by unit normal vectors away from D . Let $\mathbf{F}$ be a continuously differentiable vector field whose domain contains D . Then

$$\iint_{\partial D} \mathbf{F} \cdot d\mathbf{S} = \iiint_D \text{div } \mathbf{F} \, dV$$

Curl and divergence can be thought of as

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} \qquad \text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$$

$$\text{Where } \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

MISC

TANGENT PLANES

A plane tagent to some surface is given by

$$n_1(x - x_1) + n_2(y - y_1) + n_3(z - z_1) = 0$$

Where (n_1, n_2, n_3) is a vector normal to the surface at the point (x_1, y_1, z_1)

EIGENVALUES

A real number λ is an eigenvalue of a square matrix A if and only if $\det(A - \lambda I) = 0$, where I is the identity matrix. Example: Consider the matrix

$$A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$$

Next, setup the equation

$$\det(A - \lambda I) = 0 \quad \Rightarrow \quad \det \begin{pmatrix} 3 - \lambda & -1 \\ -1 & 2 - \lambda \end{pmatrix} = 0$$

And solve the determinant for λ .

CAUCHY-SCHWARZ

Given an inner product $\langle \cdot, \cdot \rangle$ of $\mathbb{R}^n$, we have

$$\langle \vec{v}, \vec{w} \rangle \leq \|\vec{v}\| \|\vec{w}\|$$

As such, he angle between two nonzero vectors $\vec{v}$ and $\vec{w}$ in $\mathbb{R}^n$ relative to a given inner product $\langle \cdot, \cdot \rangle$ is defined by

$$\theta = \cos^{-1} \left(\frac{\langle \vec{v}, \vec{w} \rangle}{\|\vec{v}\| \|\vec{w}\|} \right)$$

very cool